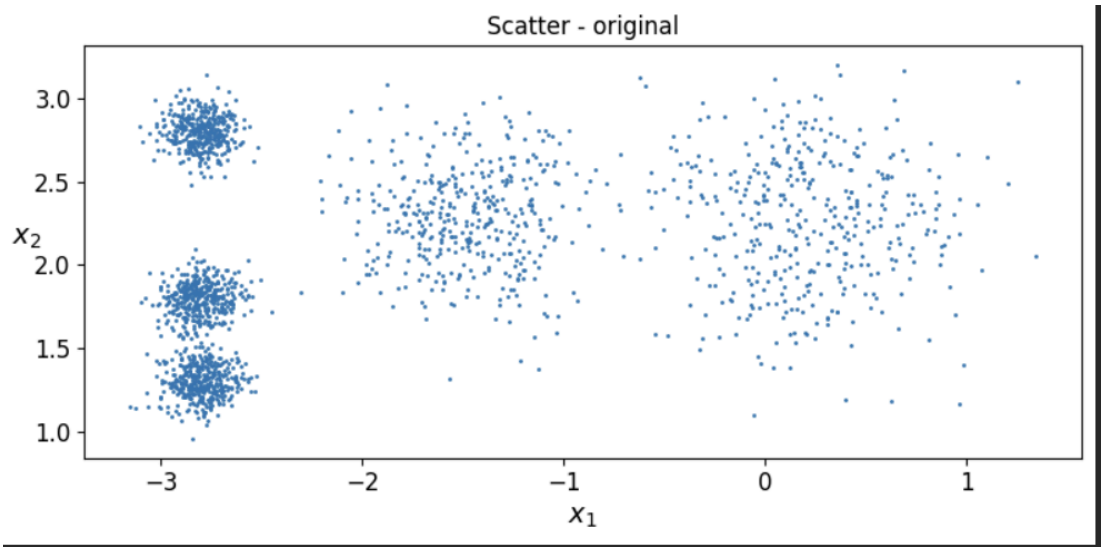
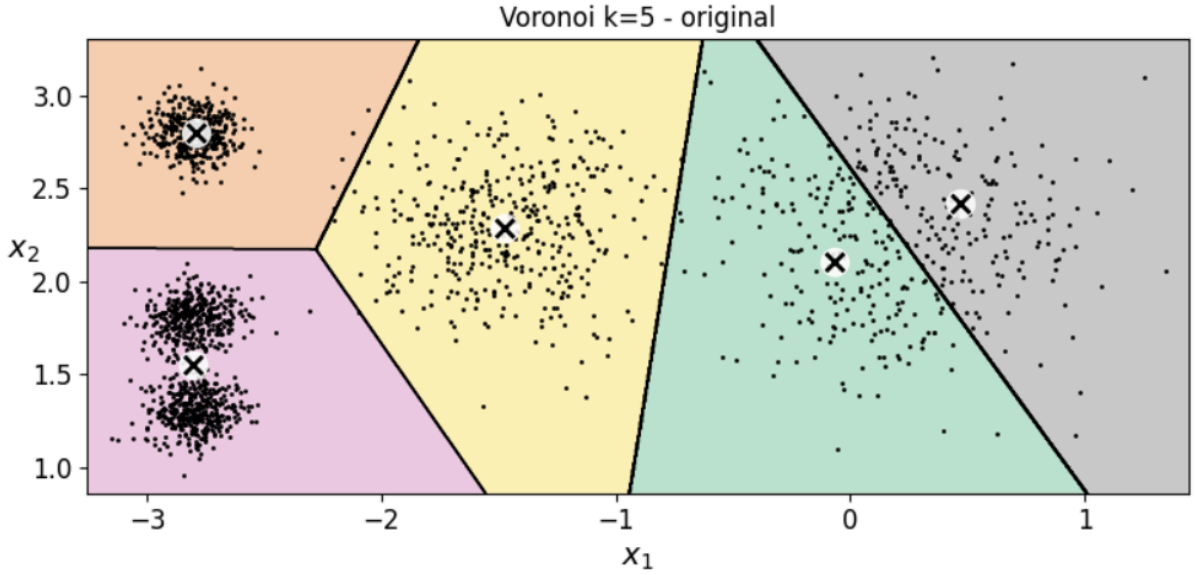
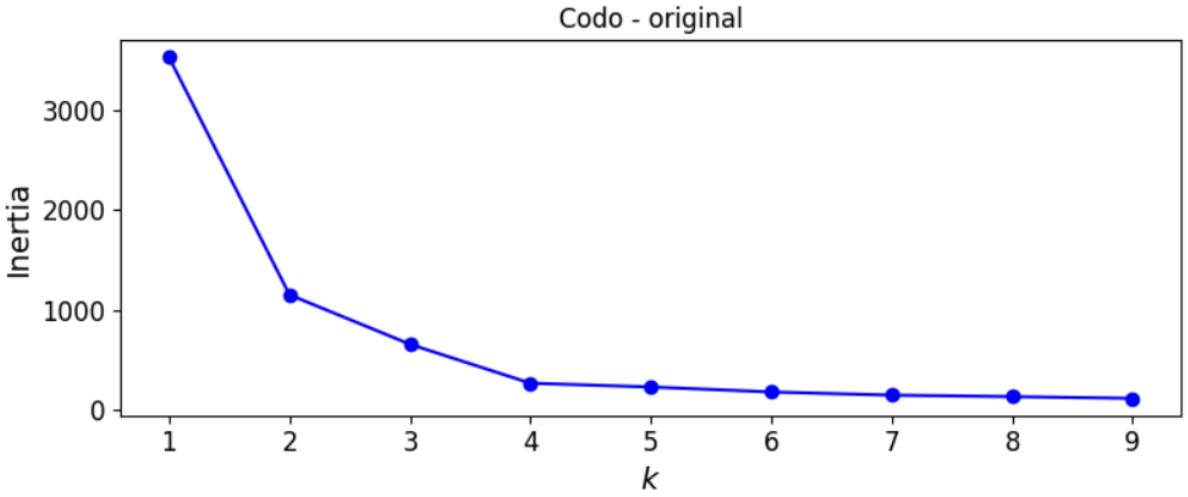
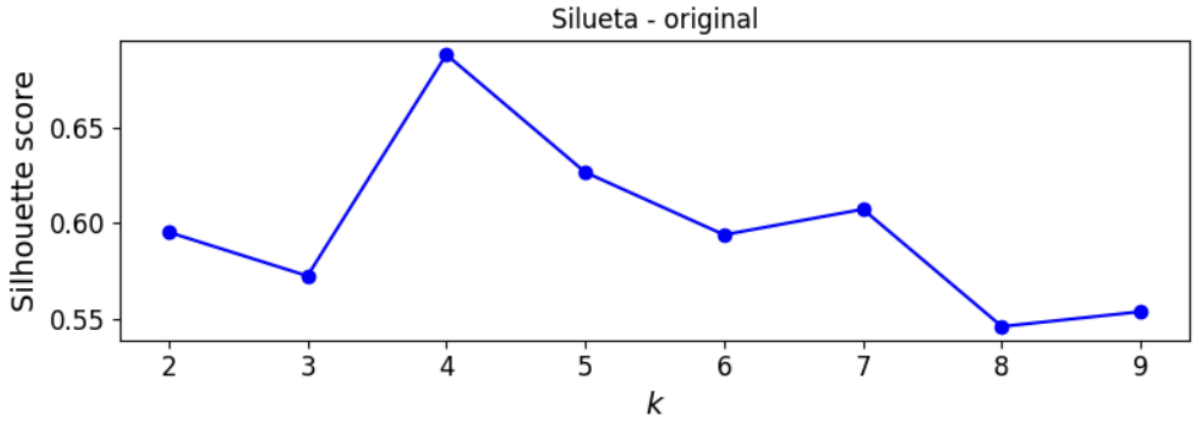
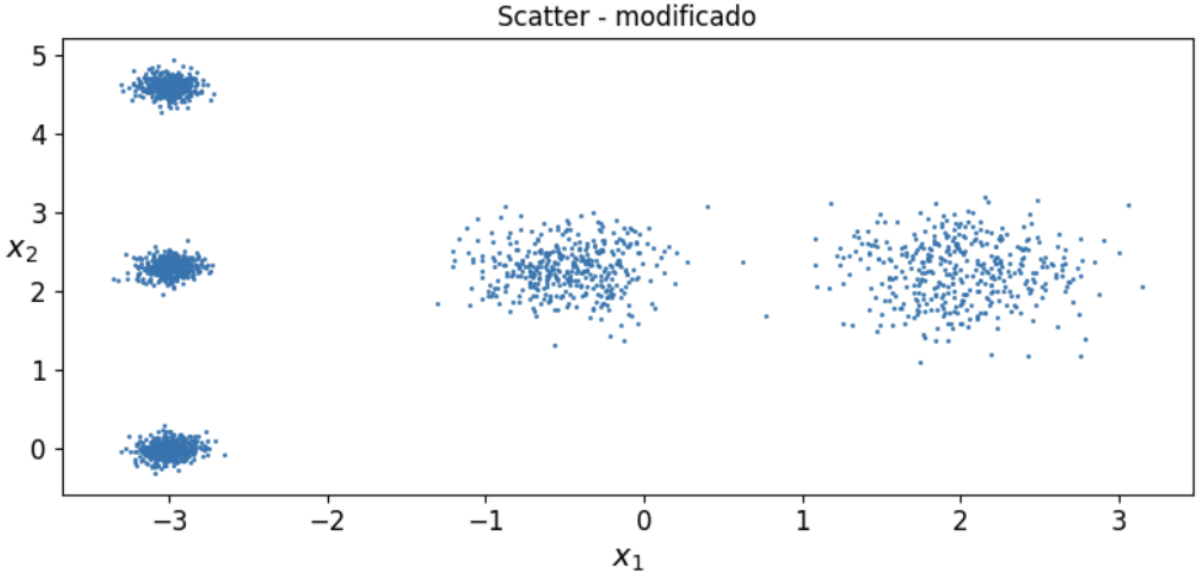
Capturas :

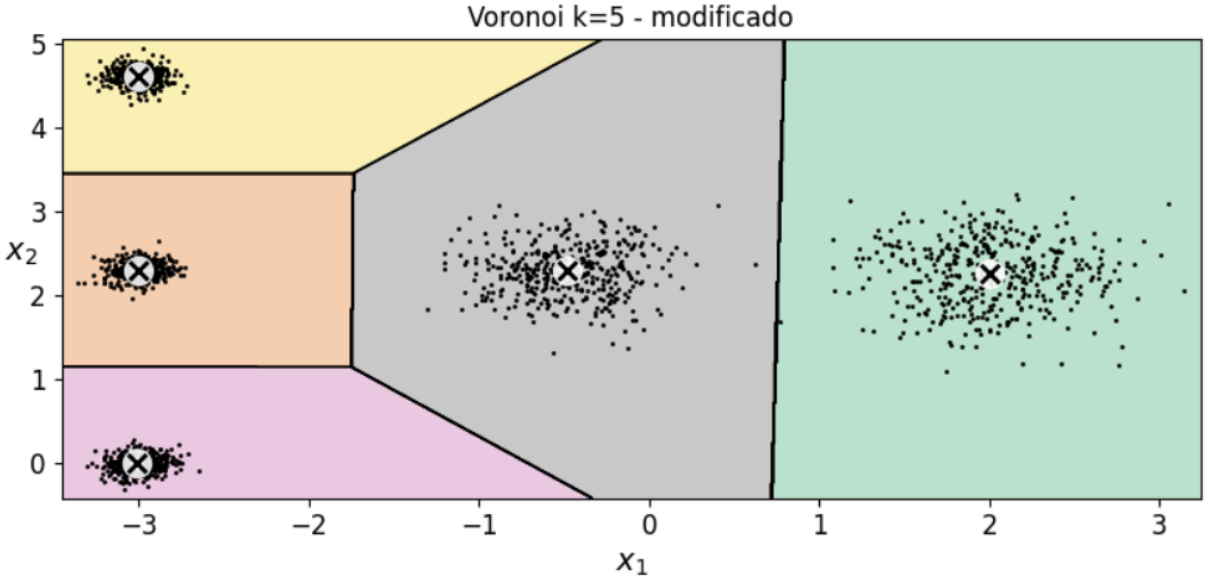

Scatter - original
3.0
2.5
2.0
1.5
1.0
x_2
−3
−2
−1
0
1
x_1

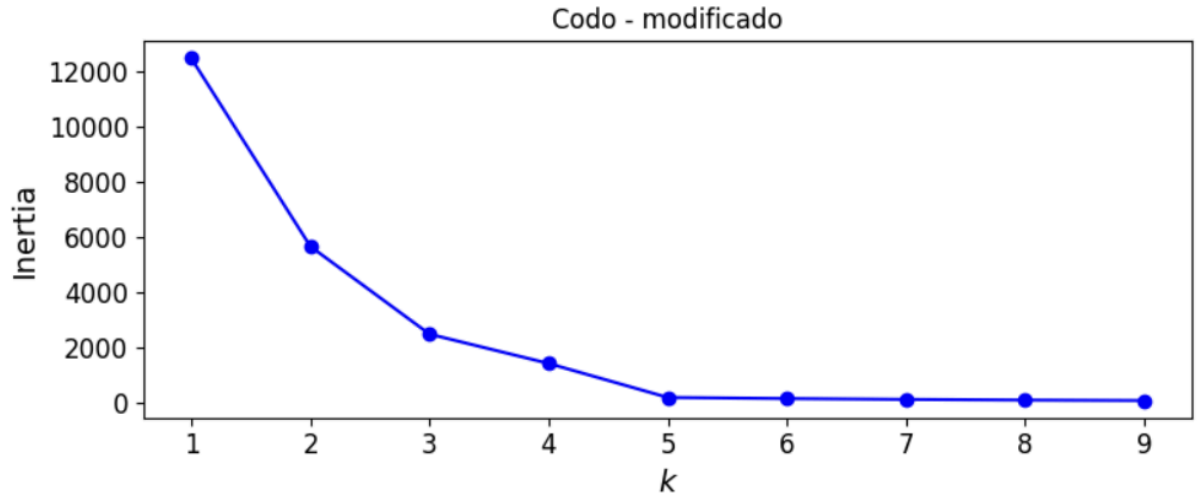

Voronoi k=5 - original
3.0
2.5
2.0
1.5
1.0
x_2
−3
−2
−1
0
1
x_1

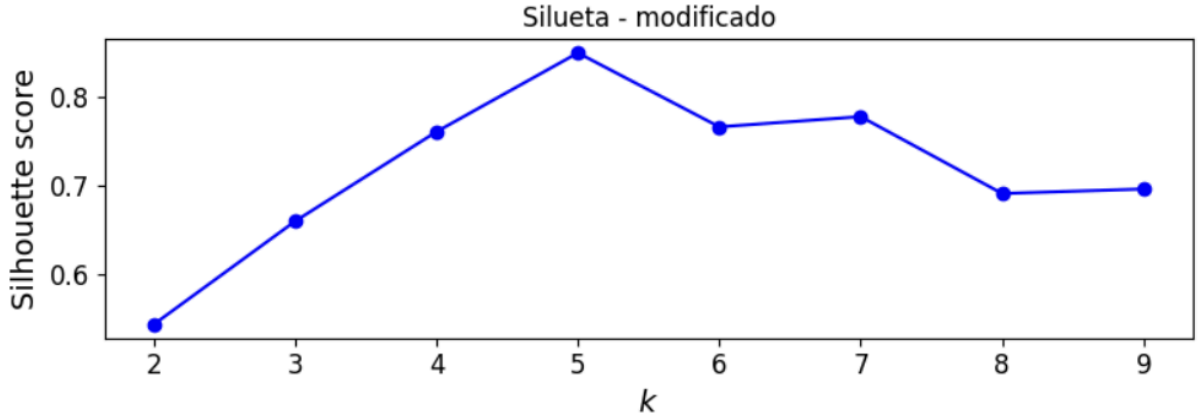

Codo - original
3000
2000
1000
0
Inertia
1
2
3
4
5
6
7
8
9
k


Silueta - original
Silhouette score
0.65
0.60
0.55
2
3
4
5
6
7
8
9
k


Scatter - modificado
5
4
3
2
1
0
x_2
−3
−2
−1
0
1
2
3
x_1


Voronoi k=5 - modificado
5
4
3
2
1
0
x_2
−3
−2
−1
0
1
2
3
x_1

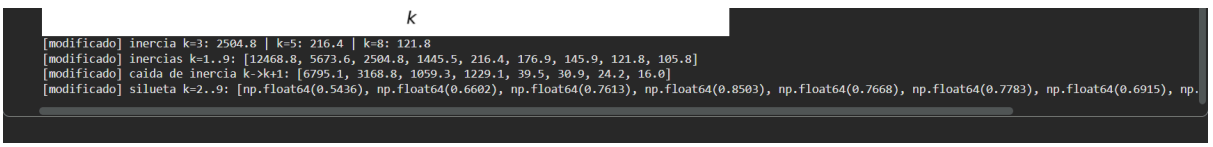




Los 5 centros y 5 std :

```
# --- CORRIDA MODIFICADA (blobs separados) ---
blob_centers_new = np.array(
    [[ 2.0,  2.3],
     [-0.5,  2.3],
     [-3.0,  0.0],
     [-3.0,  4.6],
     [-3.0,  2.3]])
blob_std_new = np.array([0.4, 0.3, 0.1, 0.1, 0.1])   # std sin cambios: solo se alejaron los centros
X_m, models_m, inert_m, sil_m = run_experiment(blob_centers_new, blob_std_new, "modificado")
```

Evidencia de haber ejecutado en Colab :



[38]
0 s

```
kmeans_per_k = [KMeans(n_clusters=k, random_state=42).fit(X)
                for k in range(1, 10)]
inertias = [model.inertia_ for model in kmeans_per_k]
```

[39]
0 s

```
plt.figure(figsize=(8, 3.5))
plt.plot(range(1, 10), inertias, "bo-")
plt.xlabel("$k$", fontsize=14)
plt.ylabel("Inertia", fontsize=14)
plt.annotate('Elbow',
             xy=(4, inertias[3]),
             xytext=(0.55, 0.55),
             textcoords='figure fraction',
             fontsize=16,
             arrowprops=dict(facecolor='black', shrink=0.1)
            )
plt.axis([1, 8.5, 0, 1300])
save_fig("inertia_vs_k_plot")
plt.show()
```

```
Saving figure inertia_vs_k_plot
```

